

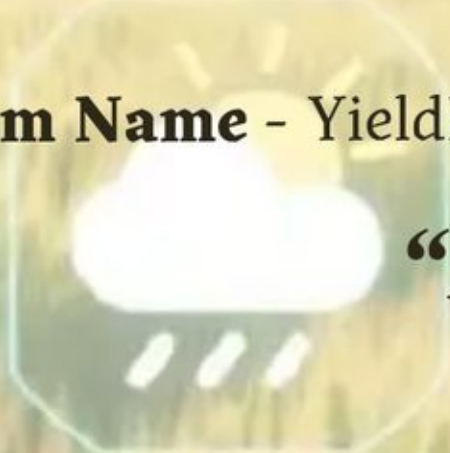
SMART INDIA HACKATHON 2025



- **Problem Statement ID** – 25044
- **Problem Statement Title**- AI-Powered Crop Yield Prediction and Optimization
- **Theme**- Agriculture, FoodTech & Rural Development
- **PS Category**- Software
- **Team ID**- 97766
- **Team Name** - YieldExperts



“Empowering Farmers with AI for Smarter Yields”



FarmIQ – AI- Powered Crop Yield Prediction & Smart Advisory

PROBLEMS

- Uncertain yields
- Real-time advisory
- Crop losses from disasters & pests
- Limited access to financial planning & govt. schemes

OUR SOLUTION

- AI driven crop yield prediction
- Smart advisory system
- Mobile app (multilingual + voice support)

Unique Innovation

- Next Crop Recommendation
- Smart Disaster Alerts
- AI Chatbot
- Profit Calculator
- Subsidy Alerts
- Offline mode

How We're Different

Feature	Existing Solutions	FarmIQ
Yield Prediction	✓ Basic Only	✓
Next Crop Recommendation	✗	✓
Disaster Alerts	✗	✓
AI Chatbot (Local Language + Voice)	✗	✓
Profit Calculator	✗	✓
Subsidy Alerts	✗	✓
All-in-One Platform	✗	✓

Crop Yield Prediction:

- 1. Programming Languages:** Python (for ML model development)
- 2. ML Frameworks:** Scikit-learn, XGBoost, TensorFlow
- 3. Data Sources:** Historical yield datasets (ICRISAT, FAO, Kaggle), soil test data, weather API data
- 4. Database:** PostgreSQL / MongoDB
- 5. Deployment:** Flask / FastAPI API

Next crop Recommendations:

- Programming Languages:** Python / JavaScript
- Rule-based System / ML Models:** Decision trees, Random Forest, or expert rules based on crop growth stage
- Data Sources:** Soil health sensors, weather forecast APIs, pest/disease databases
- Notifications:** Firebase Cloud Messaging / Twilio for alerts
- Visualization:** Dashboards using Plotly or Chart.js

Weather Data Integration

- APIs:** OpenWeatherMap, Weatherbit, or IMD API (India-specific)
- Programming Languages:** Python / Node.js to process API data
- Database:** Time-series DB like InfluxDB or normal SQL DB to store historical weather
- Data Processing:** Pandas / NumPy for analysis

Farmer Input System

- Frontend:** React.js / Flutter / Android Studio (Java/Kotlin) for mobile app
- Form Handling & Validation:** HTML5, JavaScript, or Flutter widgets
- Database:** Firebase / MongoDB / PostgreSQL

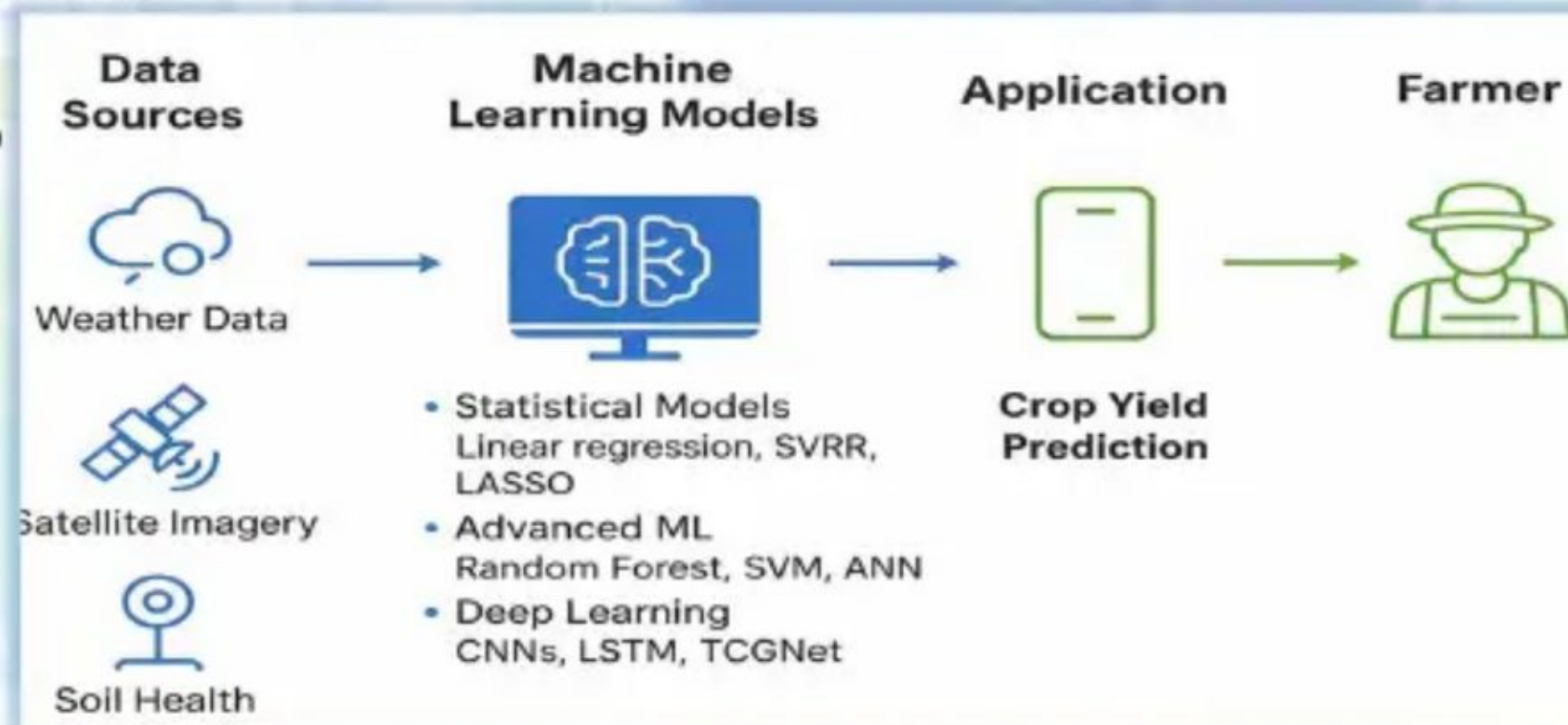
Simple Farmer App (Multilingual + Voice Input)

- Frontend Frameworks:** Flutter / React Native (cross-platform)
- Voice Input:** Google Speech-to-Text API / Whisper
- Language Support:** iNLTK, Google Translate API for local languages
- UI/UX:** Material UI / Tailwind CSS

Offline Mode

- Local Storage:** SQLite / IndexedDB
- Sync:** Background sync (service workers or Flutter offline sync)
- Conflict Handling:** Merge local and cloud data when online

TECHNICAL APPROACH



Feasibility & Viability



FarmIQ - Impact & Benefits



Research & Reference

- [1] De Clercq, Djavan, and Adam Mahdi. "Feasibility of machine learning-based rice yield prediction in India at the district level using climate reanalysis and remote sensing data." *Agricultural Systems* 220 (2024): 104099.
- [2] Bera, Asish, Debotosh Bhattacharjee, and Ondrej Krejcar. "PND-Net: plant nutrition deficiency and disease classification using graph convolutional network." *Scientific Reports* 14.1 (2024): 15537
- [3] ICRISAT. (n.d.). *Visualization* [Webpage]. Retrieved from <http://data.icrisat.org/dld/src/visualization.html>
- [4] Delaporte, A., & Bahia, K. (2022, October). *The State of Mobile Internet Connectivity 2022*. GSMA. <https://www.gsma.com/r/wp-content/uploads/2022/12/The-State-of-Mobile-Internet-Connectivity-Report-2022.pdf>
- [5] Food and Agriculture Organization (FAO). FAO. Retrieved from <https://openknowledge.fao.org/server/api/core/bitstreams/079fd83a-a510-4e7c-91af-68baf5061a9/content>
- [6] Guo, Gongde, et al. "KNN model-based approach in classification." *OTM Confederated International Conferences" On the Move to Meaningful Internet Systems"*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2003.
- [7] Sinaga, Kristina P., and Miin-Shen Yang. "Unsupervised K-means clustering algorithm." *IEEE access* 8 (2020): 80716-80727.
- [8] Roberts, David R., et al. "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure." *Ecography* 40.8 (2017): 913-929.
- [9] Torrey, Lisa, and Jude Shavlik. "Transfer learning." *Handbook of research on machine learning applications and trends: algorithms, methods, and techniques*. IGI Global Scientific Publishing, 2010. 242-264.